HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING



PROCESSING REPORT

MACHINE LEARNING IN HEALTHCARE

PROJECT: AUTOMATED CUP-TO DISC RATIO MEASUREMENT IN FUNDUS IMAGES FOR GLAUCOMA SCREENING

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Ha Noi, Oct 2026

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I. INTRODUCTION AND PROJECT OBJECTIVES

1. Clinical Context and Pathological Background

Glaucoma represents a chronic, progressive optic neuropathy characterized by the structural excavation of the optic nerve head (ONH) and the irreversible apoptosis of retinal ganglion cell axons. Because vision loss initiated by glaucomatous neuropathy cannot be medically reversed, early structural detection at asymptomatic stages is vital to prevent severe visual field constriction and permanent blindness.

Among several quantitative anatomical biomarkers derived from color fundus photography (CFP), the Cup-to-Disc Ratio (CDR)—and specifically the vertical Cup-to-Disc Ratio (vCDR)—serves as a primary clinical measurement for assessing neuroretinal rim loss. Pathological cupping leads to vertical enlargement of the optic cup relative to the overall diameter of the optic disc, causing an elevated vCDR (typically $vCDR > 0.6$). However, manual annotation and clinical estimation of the disc and cup margins by ophthalmologists are inherently subjective, time-consuming, and subject to high inter- and intra-observer variability. Automating this delineation through deep learning models offers an objective, high-throughput solution suitable for community-level screening.

2. Scope and Specific Research Objectives

The primary objective of this project is to develop and evaluate a computerized screening support pipeline based on the U-Net architecture that automatically delineates the Optic Disc (OD) and Optic Cup (OC) from digital color fundus images. The technical milestones designed for this research encompass:

- Formulating a robust data audit protocol to pair, clean, and verify raw fundus images with corresponding multi-expert ground-truth masks.
- Implementing a standardized preprocessing and multi-class target engineering pipeline that addresses extreme spatial class imbalances.
- Constructing a compact U-Net segmentation network and systematically exploring loss function variants to optimize cup boundary sensitivity.
- Selecting the optimal model checkpoint based strictly on validation performance, followed by blind evaluation on an isolated test partition.
- Establishing the foundational framework required for automated vertical diameter extraction, vCDR error estimation, and final binary glaucoma classification.

3. End-to-End Methodological Workflow

The overall execution pipeline is structured into eight sequential phases:

1. Dataset Ingestion and Sanity Auditing: Verification of the benchmark dataset to identify annotation anomalies.
2. Data Standardization and Preprocessing: Spatial resizing and multi-class label map encoding.
3. Partitioning: Splitting into training, validation, and test cohorts with preserved pathological distributions.
4. Architectural Modeling: Implementation of the 4-level U-Net model in PyTorch.
5. Systematic Loss Experimentation: Iterative model training comparing unweighted cross-entropy, weighted cross-entropy, and compound Dice loss formulations.
6. Validation Benchmarking: Identification and locking of the top-performing model checkpoint without test partition contamination.
7. Holdout Test Evaluation: Final quantitative and qualitative error analysis on unseen test images.
8. Downstream Pipeline Formulation: Defining boundary extraction algorithms for automated CDR measurement and clinical risk analysis.

II. DATASET SPECIFICATION AND DATA PREPARATION

1. The RIM-ONE DL Benchmark Dataset

This research utilizes the RIM-ONE DL (Retinal Image Database for Optic Nerve Evaluation for Deep Learning) benchmark repository. Compiled in 2020 by specialized ophthalmological teams across three clinical centers in Spain, this repository was specifically designed to evaluate machine learning algorithms for optic nerve evaluation.

The dataset comprises 485 high-resolution color fundus photographs provided in uncompressed PNG format. The cohort distribution is clinically categorized into:

- Normal (Healthy controls): 313 images.
- Glaucoma (Confirmed pathological cupping): 172 images.

Each individual image is accompanied by pixel-level reference contours for both the Optic Disc and the Optic Cup, established by clinical experts to serve as ground-truth evaluation standards.

2. Quality Verification and Anomaly Auditing

A comprehensive automated pairing script was executed across the directory hierarchy to ensure that each fundus image possessed an exact spatial match with its corresponding disc and cup mask. The audit confirmed 485 complete tripartite image-disc-cup pairs.

During topological consistency checking, an anatomical anomaly was identified in 5 distinct masks, wherein segmented Optic Cup pixels were situated partially outside the designated Optic Disc margin. From a biological standpoint, the optic cup is an excavation strictly situated inside the perimeter of the optic disc. To ensure strict data provenance and prevent subjective manual bias, the raw ground-truth masks were left entirely unedited. Instead, a systematic rule—termed the Cup Priority Rule—was integrated into the automated label generator: whenever a pixel is marked as Optic Cup, it is unconditionally assigned to the Cup class, regardless of its disc-mask containment.

3. Dataset Partitioning Protocol and Methodological Constraints

The 485 images were split into three independent partitions: a Training Set, a Validation Set, and a Test Set. The exact sample and diagnostic distributions are structured as follows:

Dataset Partition	Normal Images	Glaucoma Images	Total Images	Partition Percentage
Training Set	175	96	271	55.88%
Validation Set	44	24	68	14.02%
Test Set	94	52	146	30.10%
Total Cohort	313	172	485	100.0%

The Training Set was utilized exclusively for backpropagation and weight updates. The Validation Set guided hyperparameter adjustments, learning rate schedules, and checkpoint selection. The Test Set was strictly isolated to measure real-world performance.

Methodological Limitation: Due to the absence of explicit, de-identified patient identifiers (patient_id) in the public metadata of RIM-ONE DL, splitting was performed at the image/record level rather than the patient level. Consequently, there remains a minor risk that bilateral eye captures (OD/OS) or repeat captures from the same subject could be

distributed across different splits, presenting an external validity constraint that must be resolved in future clinical studies.

4. Image Standardization and 3-Class Semantic Target Generation

To balance computational throughput with morphological resolution, all input images and masks were standardized to a spatial dimension of 256×256 pixels:

- Fundus photographs were resized using bilinear interpolation to ensure smooth channel continuity across retinal microvascular features.
- Ground-truth masks were resized using nearest-neighbor interpolation to prevent the introduction of intermediate, non-discrete class labels along boundary borders.

Rather than optimizing two disparate binary segmentation networks, a unified 3-class segmentation ground-truth map was generated for each image:

- Class 0 (Background): Surrounding retinal tissue and non-disc anatomical regions.
- Class 1 (Disc-only): The neuroretinal rim region, defined as pixels belonging to the Optic Disc but excluding the Optic Cup.
- Class 2 (Optic Cup): The central excavated cup region, enforced via the Cup Priority Rule.

During downstream performance evaluation, full binary representations are deterministically reconstructed:

$$\text{Reconstructed Disc Mask} = (\text{Class 1}) \cup (\text{Class 2})$$

$$\text{Reconstructed Cup Mask} = \text{Class 2}$$

The fully preprocessed dataset was cataloged at /content/preprocessed_256 within the Google Collab environment.

III. MODEL ARCHITECTURE & TRAINING CONFIGURATION

1. U-Net Architectural Formulation

Semantic segmentation is executed via a customized PyTorch implementation of classic U-Net architecture. U-Net consists of an encoder (contracting path) and a decoder (expansive path):

- Contracting Path (Encoder): Four consecutive stages consisting of dual 3×3 convolutional layers followed by Batch Normalization, ReLU activations, and 2×2 max-pooling operations. This path extracts hierarchical contextual semantics while halving spatial dimensions at each stage.

- Bottleneck: A dense central layer capturing maximum receptive field representations.
- Expansive Path (Decoder): Four decoding stages employing transposed convolutions (2×2 up-convolution) to progressively reconstruct spatial dimensions.
- Lateral Skip Connections: Direct concatenation links connecting corresponding encoder feature maps to decoder inputs. These connections supply high-frequency spatial gradients, enabling the network to resolve subtle morphological edges along the disc and cup margins.
- Segmentation Head: A concluding 1×1 convolution mapping 64 feature channels directly into 3 class output logits.

The architecture maintains a parameter footprint of 482,067 trainable parameters, ensuring efficient gradient computation on consumer-tier cloud hardware while mitigating over-parameterization risks.

2. Training Environment and Hyperparameters

- Deep Learning Framework: PyTorch 2.x.
- Hardware Accelerator: NVIDIA Tesla T4 GPU (16 GB VRAM) allocated via Google Colab.
- Mini-Batch Size: 4 samples per gradient step.
- Optimization Algorithm: Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$).

3. Multi-Loss Experimental Design

Three distinct training setups were implemented to analyze the impact of loss formulation and class penalization on segmentation quality:

3.1. Experiment 1: Baseline Vanilla U-Net

- Loss Function: Standard Unweighted Multiclass Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{2} y_{i,c}\log\left(\hat{p}_{i,c}\right)$$

- Hyperparameters: Initial Learning Rate $\eta = 10^{-3}$, trained for 30 epochs.
- Objective: Establish the foundational baseline performance of the architecture without loss rebalancing.

3.2. Experiment 2: Class-Weighted Cross-Entropy

- Loss Function: Weighted Cross-Entropy Loss:

$$\mathcal{L}_{\text{WCE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{2} w_c \cdot y_{i,c} \log(\hat{p}_{i,c})$$

- Class Weight Vector: $[w_0, w_1, w_2] = [1.0, 1.0, 2.0]$.
- Hyperparameters: Initial Learning Rate $\eta = 3 \times 10^{-4}$, trained for 15 epochs.
- Objective: Double the penalty for Optic Cup prediction errors to compensate for its small pixel area relative to the retinal background.

3.3. Experiment 3: Combined Weighted Cross-Entropy and Soft Dice Loss

- Loss Function: Hybrid pixel-level and region-level objective:

$$\mathcal{L}_{\text{Combined}} = \mathcal{L}_{\text{WCE}} + \lambda \cdot \mathcal{L}_{\text{Dice}}$$

where $\lambda = 1.0$, and the multi-class Soft Dice Loss is formulated as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{C}\sum_{c=0}^{2} \frac{2\sum_{i=1}^{N} y_{i,c}\,\hat{p}_{i,c} + \epsilon}{\sum_{i=1}^{N} y_{i,c}^2 + \sum_{i=1}^{N} \hat{p}_{i,c}^2 + \epsilon}$$

- Class Weight Vector: $[w_0, w_1, w_2] = [1.0, 1.0, 2.0]$.
- Hyperparameters: Initial Learning Rate $\eta = 10^{-4}$, trained for 15 epochs.
- Objective: Optimize both individual pixel classification accuracy and macro-scale regional contour overlaps.

IV. EXPERIMENTAL RESULTS, VALIDATION BENCH MARKING, AND DISCUSSION

1. Training Dynamics and Convergence Analysis

The empirical training trajectory was documented by tracking Training Loss and Validation Loss across all epochs. Focusing on the 30-epoch training curve of Experiment 1:

- Initial Optimization Phase (Epochs 1–15): The model exhibited rapid learning; Training Loss dropped sharply from 0.92 to 0.56, and Validation Loss fell from 0.80 to 0.53. A minor gradient correction occurred on the validation curve at epoch 8 (≈ 0.74) before steady descent resumed.

- Plateau Phase (Epochs 16–24): Both loss trajectories flattened and stabilized within the range of $0.50 - 0.51$, indicating convergence toward an initial local minimum.
- Final Convergence Phase (Epochs 25–30): The loss completed a final downward transition, stabilizing between 0.42 and 0.44. The best model checkpoint was preserved at epoch 29 with an optimal validation loss of 0.42118.
- Overfitting Evaluation: The validation curve closely tracked the training curve throughout the entire 30 epochs without divergent upward drift, confirming that the network maintained stable generalizability on validation samples.

Note on Loss Comparability: The minimum validation loss values recorded for Experiment 2 (0.47198 at epoch 7) and Experiment 3 (0.94142 at epoch 14) are mathematically higher due to the introduction of non-unitary class weights and the additive Dice component. These values reflect modifications to the loss objective rather than inferior model convergence.

2. Comparative Model Benchmarking on Validation Cohort

To establish an objective comparison, all three checkpoints were evaluated on the 68 validation images using the Sørensen-Dice Similarity Coefficient (DSC):

$$DSC = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$$

Mean Dice is calculated as the arithmetic average of Disc Dice and Cup Dice:

$$\text{Mean Dice} = \frac{DSC_{\text{Disc}} + DSC_{\text{Cup}}}{2}.$$

Model Setup	Loss Function	Disc Dice	Cup Dice	Mean Dice	Checkpoint Epoch	Best Val Loss
Experiment 1	CrossEntropyLoss	0.827612	0.304672	0.566142	Epoch 29	0.42118
Experiment 2	Weighted CrossEntropy	0.839380	0.414867	0.627124	Epoch 7	0.47198
Experiment 3	Weighted CE + Dice Loss	0.878706	0.466676	0.672691	Epoch 14	0.94142

Key Empirical Insights:

2.1. Deficiencies of Vanilla Cross-Entropy

Baseline U-Net achieved an acceptable Disc Dice of 0.8276, but performed poorly on the Optic Cup (0.3047). The standard cross-entropy loss was dominated by majority background pixels, leaving the network insensitive to the subtle transitions of the inner cup margin.

2.2. Impact of Class Weighting

Doubling the loss penalty on Cup pixels (Experiment 2) raised Cup Dice from 0.3047 to 0.4149 (+11.02% absolute gain), demonstrating that class rebalancing directs gradient updates toward smaller structures.

2.3. Superiority of Compound Loss

Experiment 3 produced the highest metrics across all evaluated categories: Disc Dice (0.8787), Cup Dice (0.4667), and Mean Dice (0.6727). While weighted cross-entropy establishes smooth pixel-level gradients, the region-based Dice Loss directly optimizes boundary overlap, producing more coherent contours.

2.4. Persistent Cup Segmentation Challenge

Across all models, Cup Dice remained substantially lower than Disc Dice. This disparity stems from intrinsic clinical ambiguity: unlike the clear outer perimeter of the optic disc, the boundary of the optic cup is marked only by subtle shifts in neuroretinal tissue color and blood vessel bends (kinking).

3. Model Selection Decision

Based strictly on validation metrics, the Weighted CrossEntropy + Dice Loss (Checkpoint: Epoch 14) model was locked as the final candidate. Model selection was conducted entirely without access to the test cohort, maintaining an unbiased assessment of generalizability.

4. Independent Holdout Test Set Evaluation

The selected model was evaluated on the 146 unseen test samples. The resulting test metrics are summarized below:

Metric Evaluation Index	Test Value (N = 146)	Validation Benchmark (N = 68)	Relative Stability

Optic Disc Dice	0.8905	0.8787	+0.0118 (Slight gain)
Optic Cup Dice	0.4625	0.4667	-0.0042 (Negligible delta)
Mean Dice	0.6765	0.6727	+0.0038 (Stable)

The test outcomes confirm that the segmentation pipeline generalizes consistently to unseen fundus images. The Disc Dice score (0.8905) reflects strong localization of the outer optic nerve head. However, the lower Cup Dice (0.4625) highlights that cup boundary delineation remains a technical bottleneck.

5. Qualitative Error Analysis of Predicted Segmentations

Visual inspection of the predicted masks against the ground truth revealed several key morphological patterns:

- Optic Disc Boundaries: Disc perimeters were generally segmented cleanly and closely followed by anatomical margins.
- Excessive Disc Expansion: In a small subset of samples with surrounding peripapillary atrophy (PPA), the model tended to expand disc predictions outward into the hypopigmented crescent.
- Cup Fragmentation: The network occasionally segmented the optic cup into disconnected pixel islands rather than a unified convex shape.
- Boundary Instability: For pale fundus backgrounds, the model struggled to differentiate physiological cupping from true glaucomatous excavation, leading to irregular contours.

V. REPOSITORY STRUCTURE & EXPERIMENTAL ARTIFACTS

All training weights, metric logs, and generated visual masks have been cataloged in Google Drive at /content/drive/MyDrive/ET4248E_Glaucoma/UNet_Results. The primary deliverables include:

1. best_unet.pt: Checkpoint weights for the Baseline U-Net (Epoch 29).
2. weighted_cup_experiment/best_unet_weighted.pt: Checkpoint weights for Weighted Cross-Entropy (Epoch 7).
3. combined_loss_experiment/best_unet_combined.pt: Checkpoint weights for the final Weighted CE + Dice Loss model (Epoch 14).

4. validation_model_comparison.csv: Validation evaluation logs across all experimental setups.
5. final_test_summary.csv: Aggregated performance metrics on the 146 test samples.
6. test_per_image_dice.csv: Granular image-level Disc Dice and Cup Dice metrics for every test sample.
7. Training curve logs and diagnostic visualization plots.

VI. PROJECT MILESTONES, CURRENT LIMITATIONS, AND FUTURE WORK

1. Summary of Completed Milestones

- Fully audited and validated the 485-sample RIM-ONE DL dataset, ensuring complete ground-truth alignment and preserving outlier labels under the Cup Priority Rule.
- Formulated an end-to-end preprocessing, spatial scaling (256×256), and 3-class target representation pipeline.
- Implemented a 482,067-parameter U-Net architecture in PyTorch.
- Benchmarked three loss configurations, identifying the hybrid Weighted CE + Dice Loss as the top-performing setup.
- Evaluated generalizability on 146 test samples and conducted a qualitative visual error audit.

2. Incomplete Tasks and Methodological Limitations

While the segmentation phase is complete, several components required for the automated glaucoma screening pipeline remain pending:

- Automated CDR Computation: Algorithmic extraction of vertical cup diameter and vertical disc diameter directly from the model-predicted masks has not yet been executed.
- CDR Error Quantification: The system has not yet measured the statistical error (Mean Absolute Error - MAE, Root Mean Squared Error - RMSE, and Pearson correlation coefficients) between predicted CDRs and ground-truth reference CDRs.
- Integrated Glaucoma Classification: Downstream classification (e.g., Logistic Regression or SVM) using model-predicted CDR values has not yet been integrated into an automated end-to-end pipeline.

- Patient-Level Data Isolation: Due to metadata constraints in RIM-ONE DL, splits were organized at the image level rather than the patient level, leaving potential inter-eye correlations unaddressed.

3. Action Plan and Future Development

The subsequent phase of this project will focus on the following steps:

1. Vertical Diameter Boundary Extraction: Implement bounding-box and contour-scanning algorithms to extract vertical diameters (V_{cup} and V_{disc}) from reconstructed binary masks to calculate:

$$\text{vCDR}_{\text{pred}} = \frac{V_{\text{cup}}}{V_{\text{disc}}}$$

2. Morphological Post-Processing: Apply convex hull fitting, morphological opening/closing, and connected component analysis to resolve cup fragmentation and remove spurious outlier pixels.
3. Statistical Error Benchmarking: Compute MAE and RMSE against expert reference measurements across the 146 test samples.
4. Clinical Triage Classifier: Train a Logistic Regression model using predicted vCDR as an input feature to benchmark sensitivity, specificity, and ROC-AUC for binary glaucoma screening.

VII. CONCLUSION

This research successfully implemented and evaluated an automated U-Net pipeline for Optic Disc and Optic Cup segmentation using the RIM-ONE DL dataset. Systematic experimentation demonstrated that combining Weighted Cross-Entropy with Soft Dice Loss achieved the best validation metrics: Disc Dice $= 0.8787$, Cup Dice $= 0.4667$, and Mean Dice $= 0.6727$. On an independent holdout test set of 146 images, the selected model achieved a Disc Dice of 0.8905 and a Cup Dice of 0.4625.

The results show that while optic disc boundary extraction is well-handled by the network, segmenting the optic cup remains a technical challenge that impacts automated CDR measurement. Future work will focus on integrating morphological post-processing, computing automated vCDR error metrics, and evaluating the end-to-end glaucoma classification workflow.